INDIVIDUAL WORK ALLOCATION & DELIVERY PLAN

Multi-Dataset Outbreak Investigation & Evidence Reconstruction Workspace

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Project	OUTBREAK REWIND
Planned workload share	25%
Role	Backend Foundation & Dataset Onboarding Contributor
Architecture baseline	Revised multi-dataset architecture — 12 November 2026
Purpose	Individual ownership, tasks, deliverables, handoffs and acceptance criteria

Mission

Deliver a focused, dependable first layer for the data platform: backend scaffolding, dataset registration, safe file intake, basic schema profiling and import-status APIs. The scope is intentionally smaller than the other two team members.

Ownership boundary

Owns only backend foundation, investigation/dataset registry, secure CSV/ZIP intake, original-file metadata, basic schema profiling, mapping metadata capture and initial validation response. Does not own full dataset normalization, analytics semantics, cross-source comparison, Evidence Rewind, frontend pages or end-to-end integration leadership.

Workload protection: this role is deliberately scoped to approximately one quarter of the planned team effort. The other two members retain the larger implementation and integration responsibilities. Do not expand this role into full normalization, analytics or Evidence Rewind without an explicit team reallocation.

1. Shared non-negotiable project rules

- Keep all original uploaded files unchanged and preserve dataset version, source-row identity and provenance.
- Do not flatten datasets with different grains or metric definitions into one table; retain dataset-specific extensions.
- Missing is not zero. Preserve observed, estimated, unknown and not-applicable distinctions.
- Do not infer person-level cases, contacts or transmission chains from aggregate or outbreak-level sources.
- Every metric and evidence item must be traceable to a dataset version and source record.
- Only compare sources when geography, period, metric definition and units are compatible; surface uncertainty.
- Evidence Rewind proposes deterministic changes and impact previews; a human must approve changes.
- Treat the product as an investigation/research workspace, not a clinical or operational public-health decision system.

2. Detailed task breakdown

1. Backend foundation and registry

- Set up the initial FastAPI project structure, configuration, Pydantic request/response models and database connection.
- Create the minimum database migrations for investigation, dataset, dataset version and import job metadata.
- Implement basic create/list/detail APIs for investigations and registered datasets, with ownership checks.
- Record filename, file type, checksum, upload timestamp, source label, row count when available and import status.

2. Safe file intake

- Implement CSV upload and ZIP inspection/extraction for the agreed MVP file types.
- Enforce agreed file-size and archive extraction limits; reject unsupported formats, malformed files and unsafe archive paths.
- Store original uploaded files in the configured private storage location and retain a reference to the original file.
- Do not overwrite earlier dataset versions when a new file is uploaded.

3. Basic schema profiling and mapping metadata

- Return column names, inferred basic data types, sample rows and simple missing-value counts for review.
- Provide an API to save the user's selected source-to-canonical field mapping as metadata.
- Preserve unmapped source columns and the original column names.
- Return import progress and clear validation errors to the frontend.

4. Initial validation and handoff

- Implement a small agreed set of intake checks: empty file, unreadable file, missing header, malformed rows and unsupported file type.
- Return validation flags with severity, message and source row/column where available.
- Write endpoint examples and setup notes for Anjali Kumari and Adithya.
- Add focused tests for registry, upload, file limits, malformed input and access checks.

3. Required deliverables

- Initial FastAPI application structure and database connection.
- Minimal migrations for investigation, dataset, dataset version and import job.
- Dataset registry and import-status APIs.
- Secure CSV/ZIP intake with private original-file storage and version preservation.
- Basic schema profile, sample preview and field-mapping metadata APIs.
- Initial upload validation responses, focused backend tests and API handoff notes.

4. Definition of done and acceptance criteria

A task is complete only when the implementation is committed, documented and supported by a reproducible test, working endpoint, migration or demonstrated workflow.

- A user can register an investigation and upload more than one dataset version.
- Original uploads remain unchanged and are not overwritten by later versions.
- CSV/ZIP uploads respect documented file and extraction limits.
- Dataset metadata and import status are retrievable through documented endpoints.
- Schema profile returns source column names, basic inferred types and sample rows.
- Malformed/unsupported inputs return actionable errors and do not create a false successful import.
- Focused tests pass and the frontend owner can consume the documented endpoints.

5. Dependencies and handoffs

- To Anjali Kumari: provide stable endpoint paths and response schemas for dataset listing, upload progress, schema preview, mapping metadata and validation messages.
- To Adithya: provide dataset/version IDs, original-file references, checksums, import status and mapping metadata needed for downstream normalization and provenance.
- From Adithya: receive canonical field names, minimum metadata requirements and version identifier conventions.
- From Anjali Kumari: receive upload-screen field requirements and expected loading/error/empty states.
- Escalate work that expands into normalization, analytics or rewind instead of taking it on informally.

6. Milestones and sequencing

Sequence is dependency-based. The team should add calendar dates after confirming availability and the review deadline; no dates are invented in this plan.

Milestone	Expected contribution
M0	Agree minimal metadata schema, upload limits and endpoint contract.
M1	Deliver FastAPI skeleton, DB connection and dataset registry.
M2	Deliver CSV/ZIP intake, original-file storage and version metadata.
M3	Deliver schema preview, mapping metadata and basic intake validation.
M4	Complete focused tests, API documentation and handoff support.

7. Explicitly out of scope

- Full normalization of all nine datasets or dataset-specific epidemiological semantics.
- Surveillance analytics, chart aggregation or cross-source conflict detection.
- Evidence records, hypothesis workflow, version-diff engine or Rewind approval logic.
- Frontend pages, design system, reporting UI or overall integration ownership.
- Advanced data-quality scoring, prediction, diagnosis or person-level data modelling.

8. Review/demo contribution

Show creation of an investigation, upload of a CSV and ZIP dataset, dataset-version metadata, schema preview, mapping metadata and a malformed-file validation response.

9. Weekly progress update

- Completed: PR/commit, endpoint, migration, test result or document link.
- In progress: exact task, status and next action.
- Blocked: dependency, person who can unblock it and decision required.
- Evidence: reproducible command, test output or working demo.
- Risk/change: report scope, security or data-quality risks before implementation.

10. Collaboration and handoff checklist

- Use focused branches and pull requests; request review from affected module owners.
- Do not commit credentials, secrets or private datasets.
- Document setup and sample configuration without secrets.
- Keep field names, identifiers and status values aligned across all workstreams.
- Demonstrate from a clean checkout before marking a feature complete.

11. Shared project completion standard

- One aggregate-surveillance workflow and one outbreak/event workflow work end-to-end.
- Every displayed metric and evidence item can be traced to its dataset version and source record.
- Data limitations and unresolved warnings are visible.
- Rewind changes require human approval and preserve prior revisions.
- Known limitations are documented; mock-ups alone do not count as completed functionality.

Status note: this is a proposed work allocation derived from the revised architecture and supplied dataset inventory. It is not evidence that the listed features have been implemented, scientifically validated, load-tested or security-certified. Verify dataset permissions before deployment.